

A Project
reporton

“Fingerprint Based Attendance System using ESP32”

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CERTIFICATE

This is to certify that the Dissertation entitled “**Fingerprint Based Attendance System using ESP32**”, submitted by **Patare Ashwini , Jagtap Vaishnavi , Kodam Tanvi** is a record of bonafide work carried out by him/her, in the partial fulfillment of the requirement for the award of Third Year Degree of Bachelor of Engineering (E&TC) at DVVP College of Engineering, Vilad Ghat, Ahmednagar under the University of Pune. This work is done during year 2025-2026, under our guidance.

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Examination

Examiner 1:.....

Examiner 2:.....

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1) Patare Ashwini

2) Jagtap Vaishnavi

3) Kodam Tanvi

ABSTRACT

Attendance management is an important activity in schools, colleges, offices, and industries for maintaining proper records of students and employees. Traditional attendance systems based on manual entry are time-consuming, less secure, and may lead to errors such as duplicate entries and proxy attendance. To overcome these problems, an automated biometric attendance system can be used for accurate and reliable attendance monitoring.

This project presents the design and implementation of a **Fingerprint Based Attendance System using ESP32**. The system uses fingerprint recognition technology for identifying authorized users and recording attendance automatically. Since every individual has a unique fingerprint pattern, biometric authentication provides better security and accuracy compared to conventional attendance methods.

The main component used in this project is the ESP32 microcontroller, which offers high processing capability along with built-in Wi-Fi functionality. A fingerprint sensor module is interfaced with the ESP32 to capture and verify fingerprints. During enrollment, fingerprints of users are stored in the sensor memory with unique identification numbers. Whenever a user places a finger on the sensor, the system compares the scanned fingerprint with the stored database. If the fingerprint matches successfully, the attendance is marked automatically.

The ESP32 connects to a Wi-Fi network and sends attendance details such as user ID, date, and time to Google Sheets using Google Apps Script. This enables real-time attendance storage and easy access to attendance records from anywhere. The use of Google Sheets reduces paperwork and provides a simple method for maintaining digital attendance records. The developed system is user-friendly, reliable, and cost-effective. It reduces manual work, prevents proxy attendance, and improves the efficiency of attendance management. The project demonstrates the practical application of embedded systems, IoT technology, and biometric authentication in real-world applications.

The system can be effectively used in educational institutions, offices, laboratories, and industries where accurate attendance monitoring is required. Future enhancements such as mobile application support, cloud database integration, and advanced biometric features can further improve the system performance and functionality.

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CHAPTER

INTRODUCTION

1.1 Introduction :

In today's world, attendance management plays an important role in schools, colleges, offices, and industries. Traditional attendance systems based on manual registers are time-consuming and may lead to errors, duplicate entries, and proxy attendance. To overcome these problems, automated attendance systems using biometric technology are becoming more popular.

A fingerprint-based attendance system is one of the most reliable biometric systems because every person has a unique fingerprint pattern. The use of fingerprint recognition improves security, accuracy, and efficiency in attendance monitoring. It also reduces paperwork and manual effort required for maintaining attendance records.

This project focuses on the development of a Fingerprint Based Attendance System using ESP32. ESP32 is a powerful microcontroller with built-in Wi-Fi and Bluetooth capabilities, making it suitable for smart and efficient embedded system applications. In this system, a fingerprint sensor is used to capture and verify the fingerprint of authorized users. Once the fingerprint is matched successfully, the attendance is recorded automatically.

The system is designed to provide a simple, low-cost, and user-friendly solution for attendance management. It helps in preventing fake attendance and ensures accurate record maintenance. The project mainly involves fingerprint enrollment, fingerprint verification, attendance marking, and data storage using ESP32 programming.

This project can be effectively used in educational institutions, offices, and organizations where accurate attendance monitoring is required. Future improvements such as cloud storage, mobile application integration, and online monitoring can further enhance the system performance and usability.

1.2 Necessity :

In many schools, colleges, and offices, attendance is still recorded manually using paper or by calling roll numbers. This method is time-consuming and requires extra effort from teachers or staff. It also increases the chances of human errors, such as marking wrong attendance or missing entries. Another major problem is proxy attendance, where one person marks attendance for another, which reduces the reliability and fairness of the system.

To overcome these issues, an automated attendance system is necessary. A fingerprint-based system provides a secure and accurate solution because each person has a unique fingerprint. This makes it almost impossible to fake attendance. It also reduces manual work and saves time.

By using ESP32 and internet connectivity, attendance data can be stored in the cloud and accessed in real-time. This improves transparency and makes data management easier.

Therefore, implementing a fingerprint-based attendance system is important for improving efficiency, accuracy, and security.

1.3 Objectives :

- To develop a fingerprint-based attendance system using ESP32.
- To record attendance automatically using fingerprint recognition.
- To reduce manual work and proxy attendance.
- To provide accurate and secure attendance management.
- To store attendance records efficiently.
- To create a low-cost and easy-to-use system.

1.4 Features of System

- Easy installation and low maintenance.
- Fully automatic attendance marking.
- High accuracy and reliability in identification.
- Prevents proxy attendance and ensures security.
- Low power consumption and user-friendly operation.

CHAPTER 2

Literature survey

2.1.1 Paper 1

Title: Fingerprint Based Attendance System

Publication Year: 2021

Author: Rahul Sharma, Amit Verma

Journal Name: International Journal of Engineering Research

Summary:

This paper presents a biometric fingerprint-based attendance system designed to eliminate traditional manual attendance methods and prevent proxy attendance[1]. The system uses a fingerprint sensor to uniquely identify individuals based on their biometric characteristics. Each fingerprint is converted into a digital template and stored securely in the system database. During attendance marking, the system scans the fingerprint and matches it with stored templates to verify identity.

The proposed system improves accuracy and reduces time consumption compared to manual methods. It also minimizes human errors and ensures secure data storage. The study highlights that biometric systems are reliable and efficient for educational institutions and can be integrated with microcontrollers and cloud storage for real-time monitoring.

2.1.2 Paper 2

Title: IoT Based Smart Attendance System

Publication Year: 2022

Author: Sneha Patil, Rohan Kulkarni

Journal Name: IEEE Access

Summary:

This paper introduces an IoT-based smart attendance system that utilizes internet connectivity to transmit attendance data to a cloud platform[2]. The system uses a microcontroller integrated with sensors and WiFi modules to capture attendance and send it to a centralized database. The cloud-based storage allows real-time access and monitoring from remote locations.

The proposed solution enhances transparency and reduces manual workload. It also supports scalability, enabling multiple users and devices to connect simultaneously. The system is cost-effective and provides better data management compared to traditional attendance systems. The study emphasizes the importance of IoT in modernizing educational systems.

2.1.3 Paper 3

Title: Biometric Attendance Using ESP32

Publication Year: 2023

Author: Akash Gupta, Neha Singh

Journal Name: International Journal of Computer Applications

Summary:

This paper focuses on the implementation of a biometric attendance system using the ESP32 microcontroller[3]. The ESP32 provides built-in WiFi capabilities, making it suitable for IoT-based applications. The system captures fingerprint data using a biometric sensor and

processes it using the ESP32. The attendance data is then uploaded to an online database for storage and analysis.

The system ensures real-time data transfer and allows administrators to monitor attendance remotely. It also reduces dependency on external hardware by integrating communication features within the microcontroller. The paper concludes that ESP32-based systems are efficient, reliable, and suitable for low-cost smart attendance applications.

2.1.4 Paper 4

Title: Cloud-Based Attendance Monitoring System

Publication Year: 2021

Author: Priya Deshmukh, Kiran Joshi

Journal Name: Springer Journal

Summary:

This paper presents a cloud-based attendance monitoring system that stores attendance records in a centralized cloud database[4]. The system allows users to access attendance data from any location using web applications. It improves data security and ensures backup of records.

The study highlights the benefits of cloud computing, such as scalability, flexibility, and cost efficiency. The system can handle large volumes of data and supports multiple users simultaneously. It also provides real-time updates and analytics, making it suitable for modern educational institutions.

2.1.5 Paper 5

Title: Smart Classroom Attendance System

Publication Year: 2021

Author: Aditya Kulkarni, Rutuja Shinde

Journal Name: Journal of Smart Systems

Summary:

This paper discusses a smart classroom attendance system that integrates biometric authentication with cloud storage[5]. The system provides real-time attendance updates and allows teachers to monitor attendance through a web interface.

2.1.6 Paper 6

Title: Web-Based Attendance Dashboard

Publication Year: 2023

Author: Sagar Pawar, Meena Kale

Journal Name: International Journal of Web Engineering

Summary:

This paper focuses on the development of a web-based dashboard for attendance systems[6]. The dashboard displays real-time attendance data, including student details, subject, and time. It provides a user-friendly interface for monitoring and managing attendance records. The system uses web technologies such as HTML, CSS, and JavaScript to create an interactive UI. It allows users to view attendance data instantly and supports data visualization features. The study highlights the importance of UI design in improving user experience and system usability.

CHAPTER 3

SYSTEM OVERVIEW

3.1 BACKGROUND and CHALLENGES:

Background:

- Traditional attendance system is manual (register / roll call)
- It takes more time to mark attendance
- There are chances of proxy attendance (fake attendance)
- Data can be lost or damaged in paper records
- No real-time monitoring available

Challenges:

- Need a fast and automatic attendance system
- System should prevent proxy attendance
- Data should be stored safely
- System should work in real-time
- Should be low-cost and easy to use
- Integration of hardware and software is required
- Internet connection is needed for data transfer

3.2 Methodology

The methodology of the Fingerprint Based Attendance System using ESP32 involves the design, development, and testing of an automated attendance management system using biometric authentication. The complete process was carried out in different stages to ensure proper implementation and reliable system performance.

Initially, the required hardware components such as ESP32 microcontroller, fingerprint sensor module, connecting wires, power supply, and internet connection were arranged. The Arduino IDE software and necessary ESP32 board packages and libraries were installed for programming and communication with the fingerprint sensor.

In the first stage, the fingerprint sensor was interfaced with the ESP32 microcontroller. The sensor was configured to enroll fingerprints of authorized users. During enrollment, each fingerprint was stored in the sensor memory with a unique identification number.

After successful enrollment, the fingerprint verification process was implemented. Whenever a user places a finger on the sensor, the system scans the fingerprint and compares it with the stored fingerprint database. If the fingerprint matches successfully, the attendance is marked automatically.

The ESP32 then connects to the Wi-Fi network and sends attendance details such as user ID, date, and time to Google Sheets using Google Apps Script. This enables real-time attendance storage and easy access to records from anywhere.

The complete system was tested repeatedly to check fingerprint recognition accuracy, Wi-Fi communication, and data storage performance. Necessary debugging and corrections were performed to improve system stability and reliability.

Thus, the methodology followed in this project ensured the successful development of a secure, efficient, and automated fingerprint-based attendance system using ESP32.

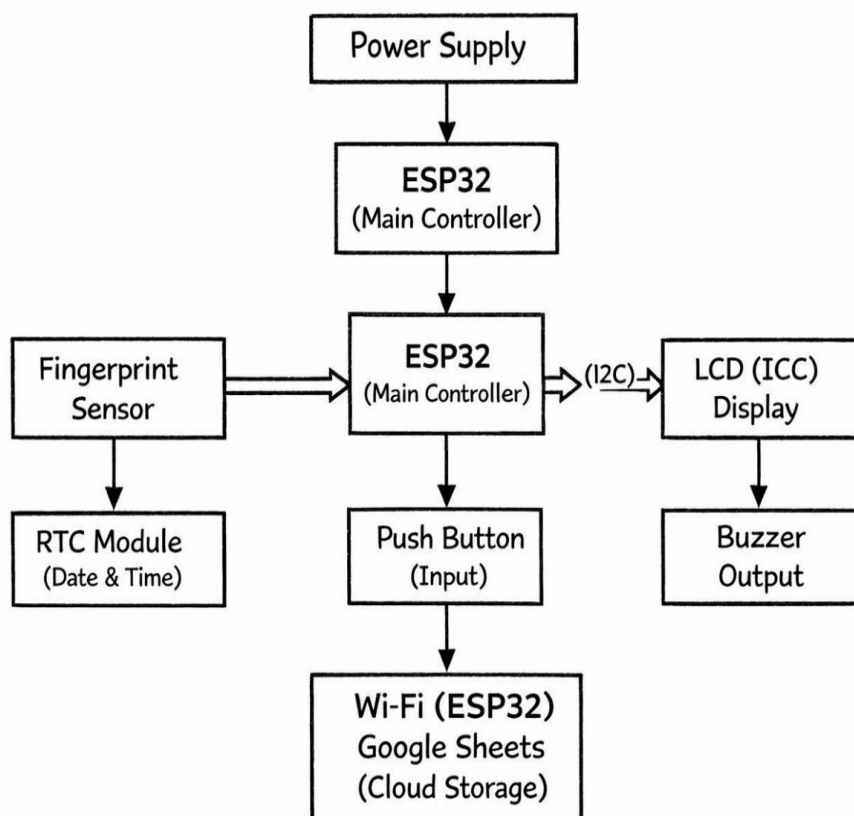
3.3 Hardware Requirements :

- Fingerprint Sensor
- ESP32
- LCD with I2C
- RTC Module
- Buzzer
- Push Buttons
- Jumper Wires
- LED

3.4 Software Requirement

- **Arduino IDE**
Used for writing and uploading code to ESP32 microcontroller.
- **Google Apps Script**
Used to create a web API for sending and receiving attendance data.
- **Google Sheets**
Used as a cloud database to store attendance records.
- **Wi-Fi Network**
- **Fingerprint Sensor Library**

3.5 BLOCK DIAGRAM :



3.6 Block Description :

1. Power Supply

The power supply provides the necessary voltage and current to operate the entire system. It typically supplies 5V or 3.3V to the ESP32 and other components like the fingerprint sensor, LCD, and RTC module. A stable power supply is important to ensure proper functioning and avoid system errors.

2. ESP32 (Main Controller)

The ESP32 acts as the central processing unit of the system. It receives input data from the fingerprint sensor and push button, processes the data, and controls all output devices such as the LCD and buzzer. It also communicates with the RTC module to fetch the current date and time. Additionally, its built-in Wi-Fi feature allows the system to send attendance data to Google Sheets for online storage and monitoring.

3. Fingerprint Sensor

The fingerprint sensor is used to capture the fingerprint image of the user. It converts the fingerprint into digital data and sends it to the ESP32 through serial communication (UART). The system compares this data with stored fingerprint templates to verify the user's identity. This ensures secure and unique identification.

4. RTC Module (Date & Time)

The Real-Time Clock (RTC) module keeps track of the current date and time continuously, even when the system is powered off (using a backup battery). It communicates with the ESP32 via I2C protocol. This module ensures that each attendance entry is recorded with accurate time and date information.

5. LCD (I2C Display)

The LCD display is used to show messages to the user. It displays information such as "Place Finger," user name, ID, attendance status (marked/denied), and date & time. The I2C interface reduces the number of connecting wires and makes communication with ESP32 simpler and more efficient.

6. Push Button (Input)

The push button acts as a manual input device. It is used for functions like enrolling a new fingerprint, deleting records, or resetting the system. When pressed, it sends a signal to the ESP32, which then performs the assigned operation.

7. Buzzer Output

The buzzer provides audio feedback to the user. It produces different sounds for different conditions, such as a short beep for successful attendance and a long or repeated beep for failed verification. This helps users understand the system response quickly without looking at the display.

8. Wi-Fi (ESP32)

The ESP32 has built-in Wi-Fi capability, which allows it to connect to the internet. After successful attendance marking, the ESP32 sends the data (user name, ID, date, and time) to an online platform. This feature enables remote monitoring and real-time updates.

9. Google Sheets

Google Sheets is used as a cloud database to store attendance records. The data sent by ESP32 is automatically updated in the sheet, making it accessible to teachers or administrators from anywhere. It helps in maintaining organized records, easy tracking, and analysis of attendance.

CHAPTER 4

SYSTEM DESIGN AND WORKING

4.1 Circuit Diagram :

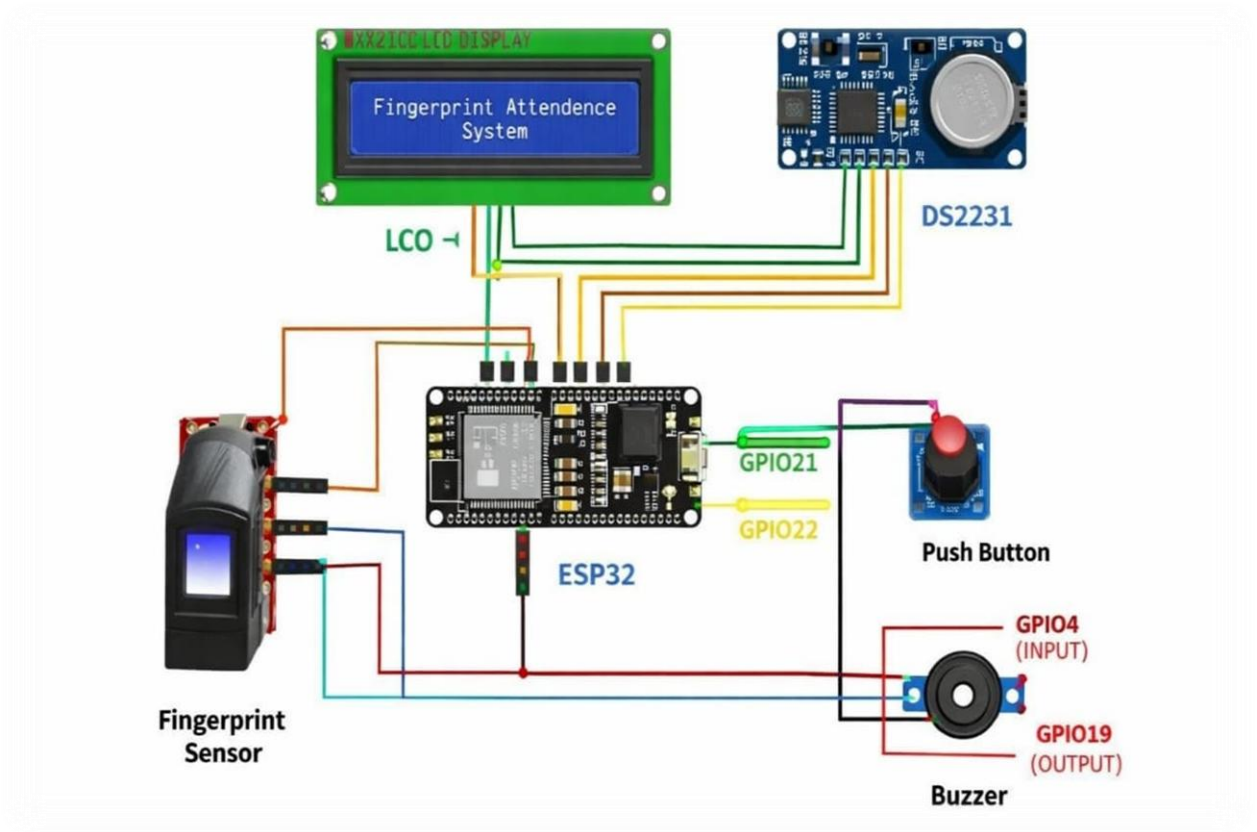


Fig. Fingerprint attendance system circuit diagram

4.2 Working :-

The working of the fingerprint-based attendance system is explained step by step as follows:

1. When the system is powered ON, ESP32 starts and connects to WiFi.
2. The LCD displays messages like "Connecting..." and "WiFi OK".
3. After initialization, the system is ready and shows the current subject, date, and time.
4. The system has two modes:
 - Enrollment Mode
 - Attendance Mode

ENROLLMENT PROCESS:

- In enrollment mode, the user selects an ID (roll number or teacher ID).
- The user places their finger on the fingerprint sensor.
- The system captures the fingerprint and asks to place the finger again.
- If both scans match, the fingerprint is stored successfully.
- The LCD displays "Stored OK".

ATTENDANCE PROCESS:

- In attendance mode, the subject is selected using buttons.
- The teacher scans their fingerprint to start attendance.
- The system activates attendance for that subject.
- Students scan their fingerprints one by one.
- The system matches the fingerprint with stored data.
- If the ID is valid:
 1. Attendance is marked
 2. Data (ID, subject, date, time) is sent to Google Sheets
- The LCD displays "Attendance Marked".
- If the same student tries again within 1 hour:
 1. System shows "Already Marked"

STOP ATTENDANCE:

- The teacher scans fingerprint again.
- The system stops attendance and shows "Attendance Closed".

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 Implementation

The implementation of the Fingerprint Based Attendance System using ESP32 was carried out in different stages including hardware setup, programming, fingerprint enrollment, attendance recording, and data storage.

Initially, the ESP32 microcontroller was connected with the fingerprint sensor module. The required libraries and ESP32 board package were installed in Arduino IDE. The fingerprint sensor was configured for enrolling and verifying fingerprints of authorized users.

During enrollment, fingerprints of users were stored in the sensor memory with unique ID numbers. After successful enrollment, the system was programmed to scan fingerprints whenever a user places a finger on the sensor.

When the fingerprint matches with the stored data, the ESP32 marks the attendance automatically. The attendance details such as user ID, date, and time are sent through Wi-Fi and stored in Google Sheets using Google Apps Script. This allows real-time attendance monitoring and easy record management.

The complete system was tested multiple times to ensure proper fingerprint recognition, stable Wi-Fi communication, and accurate attendance storage. Necessary corrections and debugging were carried out during implementation to improve system reliability and performance.

5.2 Result Analysis

The developed fingerprint-based attendance system using ESP32 was tested successfully under different conditions. The system was able to enroll fingerprints, verify authorized users, and record attendance automatically without manual intervention.

The fingerprint sensor provided accurate recognition for registered users, reducing the chances of proxy attendance and attendance errors. The ESP32 successfully connected to the Wi-Fi network and transmitted attendance data to Google Sheets in real time.

The attendance records including user ID, date, and time were stored correctly and could be viewed easily for future reference. The system worked efficiently with fast response time and user-friendly operation.

The project achieved its main objectives of providing a secure, reliable, and low-cost attendance management system. The final output demonstrated that the system can be effectively used in schools, colleges, offices, and organizations for automated attendance monitoring.

5.3 Hardware and Software Implementation

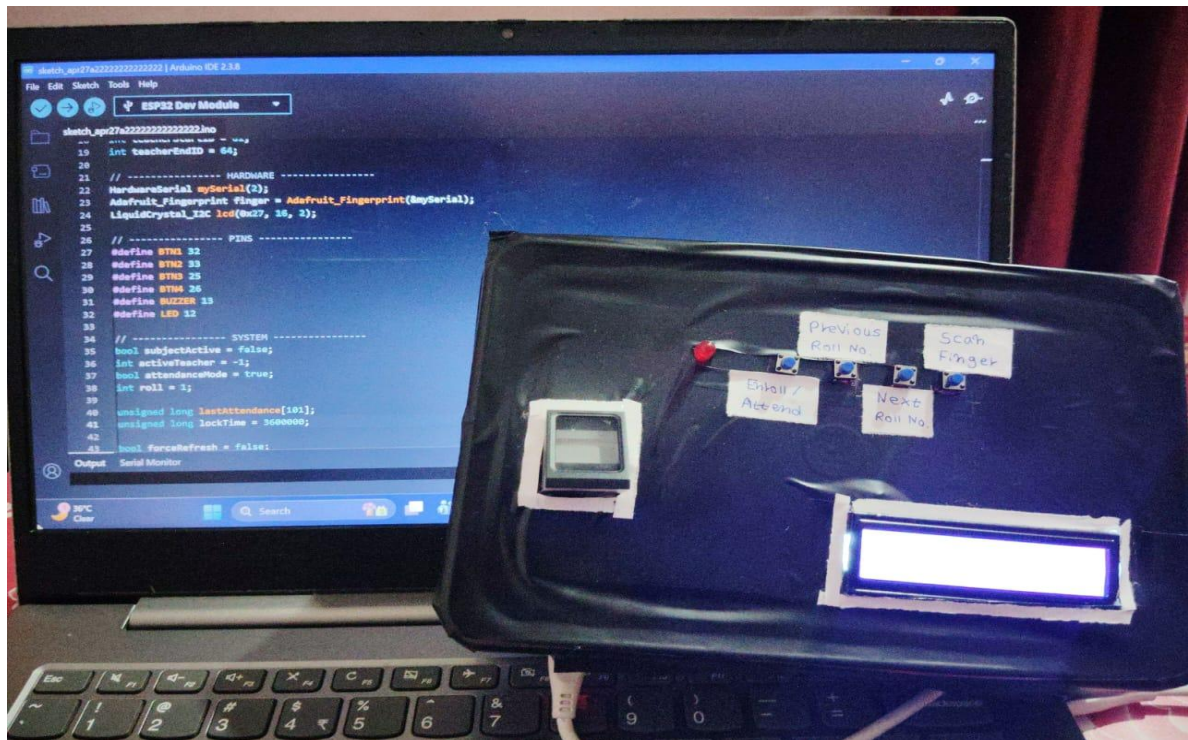


Figure 5.1: Complete Setup

The above figure shows the complete setup of the system including ESP32, fingerprint sensor, LCD display, push buttons, buzzer and Arduino IDE.

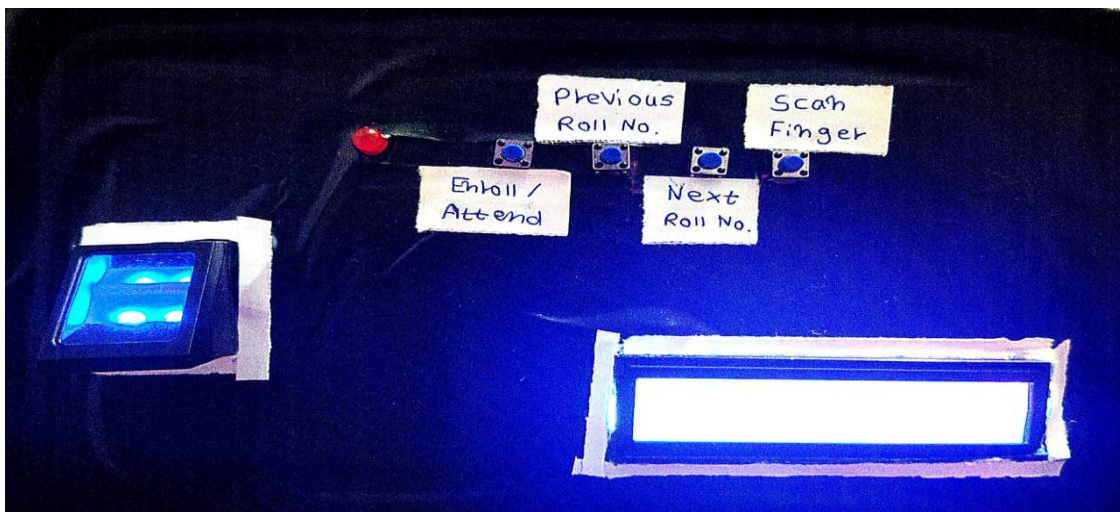


Figure 5.2: Fingerprint Scanning

This figure shows the process of scanning a fingerprint using the sensor during attendance marking.



Figure 5.3: LCD Display Output

The LCD displays messages such as "Scan Finger", "Attendance Marked", and "Already Marked".

5.4 Software Output

The screenshot shows a Google Sheet titled 'Attendance'. The table has the following data:

ROLL NO.	DATE	TIME	STATUS
2	06-05	08:44 PM	Present
24	06-05	08:45 PM	Present
24	07-05	08:00 PM	Present
2	07-05	08:01 PM	Present

Figure 5.4: Google Sheet Output

This figure shows the attendance data stored in Google Sheets including ID, date, time, and subject.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The Fingerprint Based Attendance System using ESP32 was successfully designed and developed to provide an efficient and secure method for attendance management. The project aimed to replace the traditional manual attendance system with an automated biometric-based system that improves accuracy and reliability. The system uses fingerprint authentication to identify users uniquely and mark attendance automatically without human intervention.

During the implementation process, the ESP32 microcontroller was interfaced with a fingerprint sensor module for fingerprint enrollment and verification. The ESP32 provided better processing capability and Wi-Fi connectivity, which helped in transferring attendance data directly to Google Sheets. The system was able to store attendance records along with date and time details in real time. This reduced paperwork and simplified attendance management.

The project successfully prevented proxy attendance and minimized errors commonly found in manual attendance methods. The fingerprint recognition process was accurate, fast, and user-friendly. The attendance details could be accessed easily from Google Sheets, making record maintenance simple and efficient for administrators.

The developed system is low-cost, reliable, and suitable for use in schools, colleges, offices, industries, and other organizations where attendance monitoring is important. The project also helped in understanding embedded systems, IoT applications, sensor interfacing, wireless communication, and programming using Arduino IDE.

Overall, the project achieved all its objectives and demonstrated the practical application of biometric technology in real-world attendance systems. Future improvements such as mobile application integration, cloud database support, face recognition, and online monitoring can further enhance the performance and usability of the system.

6.2 Advantages

- Eliminates proxy attendance
- Fast and accurate system
- Real-time data storage
- Easy to use
- Low cost implementation
- Secure using fingerprint authentication

6.3 Disadvantages

- Requires internet connection
- Fingerprint sensor may fail if finger is damaged or dirty
- Cannot display actual fingerprint image
- Limited storage in sensor

6.4 Applications

- Colleges and schools
- Offices and companies
- Laboratories
- Secure access systems

6.5 Future Scope

The system can be further improved by adding advanced features such as:

- Face recognition integration
- Mobile application for attendance monitoring
- Student photo display in UI
- Cloud database such as Firebase
- AI-based attendance analysis

These improvements can make the system more efficient, scalable, and suitable for real-world applications.

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